Introduction to Pandas :-

What is Pandas?

Pandas is an open-source Python library used for data manipulation, analysis, and cleaning. It provides powerful and flexible data structures that make working with structured data fast and easy.

Pandas is built on top of the NumPy library and is widely used in Data Science, Machine Learning, Data Analysis, Artificial Intelligence, and Business Intelligence.

History of Pandas :-

Pandas was created by **Wes McKinney** in **2008** while working at **AQR Capital Management**. He needed a powerful and flexible tool to analyze financial data efficiently.

Before Pandas, Python mainly relied on NumPy for numerical computing. While NumPy was excellent for numerical arrays, it lacked convenient tools for handling tabular data, missing values, and labeled data.

To solve these limitations, Wes McKinney developed Pandas. The library introduced two powerful data structures:

- **Series** – A one-dimensional labeled array.
- **DataFrame** – A two-dimensional labeled table with rows and columns.

The name **"Pandas"** is derived from **Panel Data**, an econometrics term referring to multidimensional structured datasets.

Why Pandas was Developed :-

Before Pandas, Python mainly relied on NumPy for handling numerical data. Although NumPy is fast and efficient, it was not designed to work with structured tabular data such as spreadsheets or databases.

Pandas was developed to solve these limitations by providing easy-to-use tools for data manipulation, cleaning, analysis, and handling missing values.

Problems with NumPy

- Difficult to work with rows and columns.
- No labeled rows or columns.
- Handling missing values was not convenient.
- Limited support for reading and writing files like CSV and Excel.
- Complex syntax for data analysis tasks.

How Pandas Solves These Problems

- Provides labeled data structures (Series and DataFrame).
- Makes data cleaning simple.
- Handles missing values efficiently.
- Supports reading and writing multiple file formats.

Pandas vs NumPy

Feature	NumPy	Pandas
Data Type	Numerical data	Structured & tabular data
Speed	Fast	Fast
Labels	NO	YES
Missing Value Handling	Limited	YES
CSV/Excel Support	Limited	YES
Data Analysis	Basic	Excellent

Features of Pandas :-

Key Features of Pandas

- Provides powerful data structures (Series and DataFrame).
- Fast and efficient data manipulation.
- Easy handling of missing values.
- Supports labeled rows and columns.
- Reads and writes multiple file formats (CSV, Excel, JSON, SQL, etc.).

- Powerful indexing and selection.
- Easy filtering, sorting, and grouping.
- Built-in statistical functions.

Application of Pandas :-

Pandas is widely used in various industries because it makes working with structured data simple and efficient.

Major Applications of Pandas

- 📊 Data Analysis
- 🤖 Machine Learning
- 📈 Data Science
- 💹 Financial Analysis
- 🏦 Banking and Insurance
- 🛒 Business Intelligence
- 📋 Data Cleaning and Preprocessing
- 🌐 Web Data Analysis
- 🏥 Healthcare Analytics
- 🔬 Research and Scientific Computing
- 📦 Sales and Marketing Analysis
- 📉 Stock Market Analysis
- 🗄️ Database Management
- 📑 Report Generation

Installing and Importing Pandas :-

Before using Pandas, it must be installed on your system. If you are using **Google Colab**, Pandas is already installed by default.

Install Pandas

Syntax :-

pip install pandas

Upgrade Pandas

Synatx :-

pip install --upgrade pandas

Import Pandas

Synatx :-

import pandas as pd

Using **pd** as an alias makes the code shorter and is the standard convention used by Python developers.

```
import pandas as pd

print("Pandas Version:", pd.__version__)
print("Pandas has been imported successfully!")
```

```
Pandas Version: 2.2.2
Pandas has been imported successfully!
```

Pandas Data Structures (Series and DataFrame) :-

Pandas Data Structures -

Pandas provides two powerful data structures for storing and working with data.

1. Series

A **Series** is a one-dimensional labeled array that can store data of any type, such as integers, strings, or floating-point numbers. Each value in a Series has an associated index.

Example

Index	Value
0	10
1	20
2	30
3	40

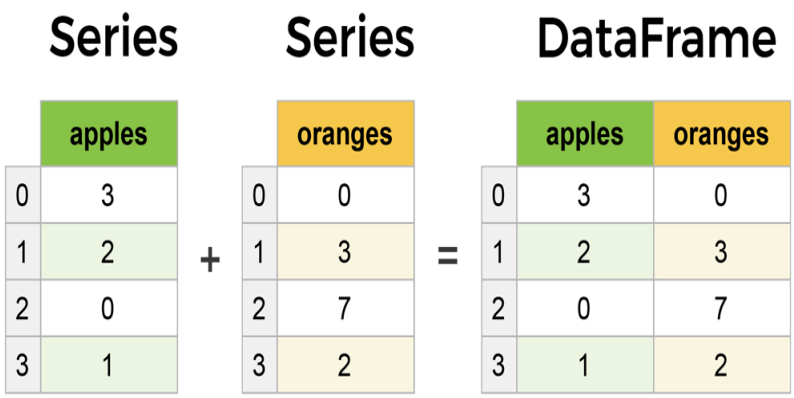
---

2. DataFrame

A **DataFrame** is a two-dimensional labeled data structure with rows and columns, similar to an Excel spreadsheet or a database table. Each column can store a different type of data.

Example

Name	Age	City
Alice	22	Mumbai
Bob	25	Pune
Charlie	24	Delhi



Creating a Series :-

A **Series** is a one-dimensional labeled array in Pandas. It can store different types of data, such as integers, floats, strings, and even Python objects.

A Series consists of two parts:

- **Index** – Identifies each element.
- **Value** – The actual data stored.

Syntax :-

pd.Series(data, index)

Parameters :-

- **data** : The values to store in the Series.
- **index** *(optional)* : Labels for each value. If not provided, Pandas automatically assigns numeric indexes starting from 0.

Create a Series from a List :-

```
import pandas as pd

s = pd.Series([10, 20, 30, 40, 50])

print(s)
```

```
0    10
1    20
2    30
3    40
```

```
4    50
dtype: int64
```

Create a Series with Custom Index :-

```
import pandas as pd

s = pd.Series([10, 20, 30], index=["A", "B", "C"])

print(s)
```

```
A    10
B    20
C    30
dtype: int64
```

Create a Series from a Dictionary :-

```
import pandas as pd

student = {
    "Name": "Alice",
    "Age": 22,
    "City": "Mumbai"
}

s = pd.Series(student)

print(s)
```

```
Name     Alice
Age         22
City    Mumbai
dtype: object
```

Accessing Elements in a Series :-

```
import pandas as pd

s = pd.Series([10, 20, 30, 40])

print(s[0])
print(s[2])
```

```
10
30
```

Creating a DataFrame :-

A **DataFrame** is a two-dimensional labeled data structure in Pandas. It stores data in rows and columns, similar to an Excel spreadsheet or a SQL table.

A DataFrame can contain different types of data in different columns.

Syntax :-

pd.DataFrame(data)

Parameters :-

bold text

- **data** : The data used to create the DataFrame. It can be a dictionary, list, list of dictionaries, NumPy array, or another DataFrame.

Create a DataFrame from a Dictionary :-

```
import pandas as pd

data = {
    "Name": ["Alice", "Bob", "Charlie"],
    "Age": [22, 25, 24],
    "City": ["Mumbai", "Pune", "Delhi"]
}

df = pd.DataFrame(data)

print(df)
```

```
      Name  Age    City
0    Alice   22  Mumbai
1      Bob   25    Pune
2  Charlie   24   Delhi
```

Create a DataFrame from a List of Dictionaries :-

```
import pandas as pd

data = [
    {"Name": "Alice", "Age": 22, "City": "Mumbai"},
    {"Name": "Bob", "Age": 25, "City": "Pune"},
    {"Name": "Charlie", "Age": 24, "City": "Delhi"}
]

df = pd.DataFrame(data)

print(df)
```

```
      Name  Age    City
0    Alice   22  Mumbai
```

```
1      Bob   25    Pune
2  Charlie   24   Delhi
```

Display Specific Columns :-

```
import pandas as pd

data = {
    "Name": ["Alice", "Bob", "Charlie"],
    "Age": [22, 25, 24],
    "City": ["Mumbai", "Pune", "Delhi"]
}

df = pd.DataFrame(data)

print(df["Name"])
print(df["Age"])
```

```
0      Alice
1        Bob
2    Charlie
Name: Name, dtype: object
0    22
1    25
2    24
Name: Age, dtype: int64
```

Reading Data :-

Pandas provides built-in functions to read data from different file formats such as CSV, Excel, JSON, SQL databases, and more.

The most commonly used functions are:

- **read_csv()** – Reads data from a CSV file.
- **read_excel()** – Reads data from an Excel file.

1. read_csv()

Reads data from a CSV (Comma-Separated Values) file and returns a DataFrame.

Syntax :-

pd.read_csv("file_name.csv")

---

2. read_excel()

Reads data from an Excel (.xlsx) file and returns a DataFrame.

Syntax :-

pd.read_excel("file_name.xlsx")

Creating a CSV File :-

```
import pandas as pd

students = {
    "ID": [101, 102, 103, 104, 105,106],
    "Name": ["Alice", "Bob", "Charlie", "David", "Emma","Alex"],
    "Age": [20, 21, 19, 22, 20,24],
    "Course": ["Python", "Java", "C++", "Python", "Java","Ruby"],
    "Marks": [85, 90, 78, 92, 88,95]
}

df = pd.DataFrame(students)

df.to_csv("students.csv", index=False)

print("students.csv created successfully!")
```

```
students.csv created successfully!
```

Reading a CSV File :-

```
import pandas as pd

df = pd.read_csv("students.csv")

print(df)
```

```
    ID     Name  Age  Course  Marks
0  101    Alice   20  Python     85
1  102      Bob   21    Java     90
2  103  Charlie   19     C++     78
3  104    David   22  Python     92
4  105     Emma   20    Java     88
5  106     Alex   24    Ruby     95
```

Read Only the First 3 Rows :-

```
import pandas as pd

df = pd.read_csv("students.csv",nrows=3)

print(df.head())
```

```
    ID     Name  Age  Course  Marks
0  101    Alice   20  Python     85
1  102      Bob   21    Java     90
2  103  Charlie   19     C++     78
```

Viewing Data :-

After loading a dataset, it is important to inspect its contents. Pandas provides several built-in functions to quickly understand the structure and summary of the data.

The most commonly used functions are:

- **head()** – Displays the first few rows.
- **tail()** – Displays the last few rows.
- **info()** – Displays information about the DataFrame.
- **describe()** – Displays statistical summary of numerical columns.

head() :-

Description: Displays the first 5 rows of the DataFrame by default.

Example :-

```
import pandas as pd

df = pd.read_csv("students.csv")

print(df.head())
```

```
    ID     Name  Age  Course  Marks
0  101    Alice   20  Python     85
1  102      Bob   21    Java     90
2  103  Charlie   19     C++     78
3  104    David   22  Python     92
4  105     Emma   20    Java     88
```

head(n) :-

Description: Displays the first n rows.

Example :-**bold text**

```
print(df.head(3))
```

```
    ID     Name  Age  Course  Marks
0  101    Alice   20  Python     85
1  102      Bob   21    Java     90
2  103  Charlie   19     C++     78
```

tail() :-

Description: Displays the last 5 rows of the DataFrame by default.

Example :-

```
print(df.tail())
```

```
   ID     Name  Age  Course  Marks
1  102      Bob   21    Java     90
2  103  Charlie   19     C++     78
3  104    David   22  Python     92
4  105     Emma   20    Java     88
5  106     Alex   24    Ruby     95
```

info() :-

Description: Displays information about the DataFrame, including column names, data types, non-null values, and memory usage.

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6 entries, 0 to 5
Data columns (total 5 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   ID      6 non-null      int64
 1   Name    6 non-null      object
 2   Age     6 non-null      int64
 3   Course  6 non-null      object
 4   Marks   6 non-null      int64
dtypes: int64(3), object(2)
memory usage: 372.0+ bytes
```

describe() :-

Description: Displays statistical information for numerical columns.

```
print(df.describe())
```

```
              ID        Age      Marks
count    6.000000   6.000000   6.000000
mean   103.500000  21.000000  88.000000
std      1.870829   1.788854   5.966574
min    101.000000  19.000000  78.000000
25%    102.250000  20.000000  85.750000
50%    103.500000  20.500000  89.000000
75%    104.750000  21.750000  91.500000
max    106.000000  24.000000  95.000000
```

Selecting Data :-

Selecting data is one of the most common operations in Pandas. You can select columns, rows, or specific values using different methods.

The most commonly used methods are:

- Selecting Columns
- Selecting Rows
- `loc[]`
- `iloc[]`

Select a Single Column :-

Description: Selects one column from the DataFrame.

```
print(df["Name"])
```

```
0      Alice
1        Bob
2    Charlie
3      David
4       Emma
5       Alex
Name: Name, dtype: object
```

Select Multiple Columns :-

Description: Selects multiple columns.

```
print(df[["Name", "Marks"]])
```

```
      Name  Marks
0    Alice     85
1      Bob     90
2  Charlie     78
3    David     92
4     Emma     88
5     Alex     95
```

Select Rows using loc[]

Description: Selects rows using their index labels.

```
print(df.loc[0])

print(df.loc[2])
```

```
ID          101
Name       Alice
Age           20
Course    Python
Marks         85
Name: 0, dtype: object
ID            103
Name      Charlie
Age            19
Course        C++
```

```
Marks          78
Name: 2, dtype: object
```

Select Rows and Columns using loc[] :-

```
print(df.iloc[0:3, 1:4])
```

```
      Name  Age  Course
0    Alice   20  Python
1      Bob   21    Java
2  Charlie   19     C++
```

Data Cleaning :-

Real-world datasets often contain missing values, duplicate records, or inconsistent data. Pandas provides several built-in functions to clean and prepare data before analysis.

The most commonly used functions are:

- `isnull()`
- `notnull()`
- `fillna()`
- `dropna()`
- `duplicated()`
- `drop_duplicates()`

```
import pandas as pd
import numpy as np

data = {
    "Name": ["Alice", "Bob", "Charlie", "David", np.nan],
    "Age": [20, np.nan, 19, 22, 21],
    "Marks": [85, 90, np.nan, 92, 88]
}

df = pd.DataFrame(data)

print(df)
```

```
      Name   Age  Marks
0    Alice  20.0   85.0
1      Bob   NaN   90.0
2  Charlie  19.0    NaN
3    David  22.0   92.0
4      NaN  21.0   88.0
```

isnull() :-

Description: Checks for missing (NaN) values.

```
print(df.isnull())
```

```
    Name    Age  Marks
0  False  False  False
1  False   True  False
2  False  False   True
3  False  False  False
4   True  False  False
```

notnull() :-

Description: Checks for non-missing values.

```
print(df.notnull())
```

```
    Name    Age  Marks
0   True   True   True
1   True  False   True
2   True   True  False
3   True   True   True
4  False   True   True
```

fillna() :-

Description: Replaces missing values with a specified value

```
print(df.fillna(0))
```

```
      Name   Age  Marks
0    Alice  20.0   85.0
1      Bob   0.0   90.0
2  Charlie  19.0    0.0
3    David  22.0   92.0
4        0  21.0   88.0
```

dropna() :-

Description: Removes rows containing missing values.

```
print(df.dropna())
```

```
    Name   Age  Marks
0  Alice  20.0   85.0
3  David  22.0   92.0
```

duplicated() :-

Description: Identifies duplicate rows.

```
print(df.duplicated())
```

```
0    False
1    False
2    False
3    False
4    False
dtype: bool
```

Filtering Data :-

Filtering is used to select rows that satisfy a specific condition. It helps you analyze only the data you need.

The most commonly used filtering methods are:

- Comparison Operators (`==`, `!=`, `>`, `<`, `>=`, `<=`)
- Multiple Conditions (`&`, `|`)
- `isin()`

Filter Using >

Description: Display students whose marks are greater than 85.

```
print(df[df["Marks"] > 85])
```

```
    Name   Age  Marks
1    Bob   NaN   90.0
3  David  22.0   92.0
4    NaN  21.0   88.0
```

Filter Using ==

Description: Display students enrolled in the Python course.

```
print(df[df["Course"] == "Python"])
```

```
    ID   Name  Age  Course  Marks
0  101  Alice   20  Python     85
3  104  David   22  Python     92
```

Multiple Conditions (&)

Description: Display students whose course is Python and marks are greater than 85.

```
print(df[(df["Course"] == "Python") & (df["Marks"] > 85)])
```

```
    ID   Name  Age  Course  Marks
3  104  David   22  Python     92
```

Multiple Conditions (|)

Description: Display students who are in the Python course or have marks greater than 90.

```
print(df[(df["Course"] == "Python") | (df["Marks"] > 90)])
```

```
   ID   Name  Age  Course  Marks
0  101  Alice   20  Python     85
3  104  David   22  Python     92
5  106   Alex   24    Ruby     95
```

Using isin()

Description: Display students enrolled in either Python or Java.

```
print(df[df["Course"].isin(["Python", "Java"])])
```

```
   ID   Name  Age  Course  Marks
0  101  Alice   20  Python     85
1  102    Bob   21    Java     90
3  104  David   22  Python     92
4  105   Emma   20    Java     88
```

GroupBy :-

The `groupby()` function is used to group rows that have the same value in one or more columns. After grouping, you can perform operations like calculating the sum, average, count, and more.

It is widely used for data analysis and reporting.

Syntax

```
df.groupby("column_name")
```

Group by Course

Description: Groups students based on the Course column.

```
group = df.groupby("Course")

print(group)
```

```
<pandas.core.groupby.generic.DataFrameGroupBy object at 0x79ba271a3a10>
```

Calculate Average Marks :-

Description: Calculates the average marks for each course.

```
print(df.groupby("Course")["Marks"].mean())
```

```
Course
C++       78.0
Java      89.0
Python    88.5
Ruby      95.0
Name: Marks, dtype: float64
```

Calculate Total Marks

Description: Calculates the total marks for each course.

```
print(df.groupby("Course")["Marks"].sum())
```

```
Course
C++        78
Java      178
Python    177
Ruby       95
Name: Marks, dtype: int64
```

Count Students

Description: Counts the number of students in each course.

```
print(df.groupby("Course")["Name"].count())
```

```
Course
C++       1
Java      2
Python    2
Ruby      1
Name: Name, dtype: int64
```

Merge & Concatenate :-

Pandas provides functions to combine multiple DataFrames.

- `merge()` – Combines DataFrames using a common column.
- `concat()` – Stacks DataFrames row-wise or column-wise.

```
import pandas as pd

students = pd.DataFrame({
    "ID": [101, 102, 103],
    "Name": ["Alice", "Bob", "Charlie"]
})
```

```
marks = pd.DataFrame({
    "ID": [101, 102, 103],
    "Marks": [85, 90, 78]
})

print(pd.merge(students, marks, on="ID"))
```

```
    ID     Name  Marks
0  101    Alice     85
1  102      Bob     90
2  103  Charlie     78
```

concat() :-

```
df1 = pd.DataFrame({"Name": ["Alice", "Bob"]})
df2 = pd.DataFrame({"Name": ["Charlie", "David"]})

print(pd.concat([df1, df2], ignore_index=True))
```

```
      Name
0    Alice
1      Bob
2  Charlie
3    David
```

Statistical Functions :-

Pandas provides built-in functions for statistical analysis.

- `sum()`
- `mean()`
- `min()`
- `max()`
- `count()`
- `std()`

```
print(df["Marks"].sum())
print(df["Marks"].mean())
print(df["Marks"].min())
print(df["Marks"].max())
print(df["Marks"].count())
print(df["Marks"].std())
```

```
528
88.0
78
95
6
5.966573556070519
```

Date & Time :-

The to_datetime() function converts dates into Pandas datetime format, allowing you to extract components like year, month, and day.

```
import pandas as pd

dates = pd.DataFrame({
    "Date": ["2026-01-15", "2026-02-20", "2026-03-10"]
})

dates["Date"] = pd.to_datetime(dates["Date"])

print(dates)

print(dates["Date"].dt.year)
print(dates["Date"].dt.month)
print(dates["Date"].dt.day)
```

```
        Date
0 2026-01-15
1 2026-02-20
2 2026-03-10
0    2026
1    2026
2    2026
Name: Date, dtype: int32
0    1
1    2
2    3
Name: Date, dtype: int32
0    15
1    20
2    10
Name: Date, dtype: int32
```

Exporting Data :-

Pandas allows you to save DataFrames to different file formats.

- `to_csv()`
- `to_excel()`

```
df.to_csv("output.csv", index=False)

df.to_excel("output.xlsx", index=False)
```

Conclusion :-

Pandas is a super helpful Python tool that acts like a powerful, automated version of Excel. It helps you open, read, and organize large amounts of messy data into neat tables called DataFrames. With Pandas, you can easily clean up missing information, filter out what you do not need, and sort your data in just a few seconds.

Ultimately, Pandas saves time by doing the heavy lifting of data preparation. It easily connects with other Python tools used for making charts or building smart AI models. It is the essential first step for anyone wanting to turn raw, confusing numbers into clear, useful facts.